



Let the mind manage the body
Que l'esprit gère le corps

SCIENCE

(Subject code No. P141/1)

Index Number:

MAURITIUS EXAMINATIONS SYNDICATE
Primary School Achievement Certificate Assessment
Grade 5 Modular
October 2017

Time: 1 hour

Total Marks: 50

INSTRUCTIONS TO CANDIDATES

1. Check that this assessment booklet contains 6 questions printed on 13 pages numbered 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14.
2. Write your Index Number on the assessment booklet in the space provided above.
3. You should not use red, green or black ink in answering questions.
4. Write all your answers clearly in the assessment booklet.
5. Attempt all questions.

Question	Marking		Revision		Control	
	Marks	Sig	Marks	Sig	Marks	Sig
1						
2						
3						
4						
5						
6						
Total						
Sig (HoG)						

Question 1 (5 marks)

Circle the letter which shows the correct answer.

1. **Diagram 1** shows a plant.

Which part of the plant holds the **seeds**?

- A Flower
- B Fruit
- C Leaf
- D Stem

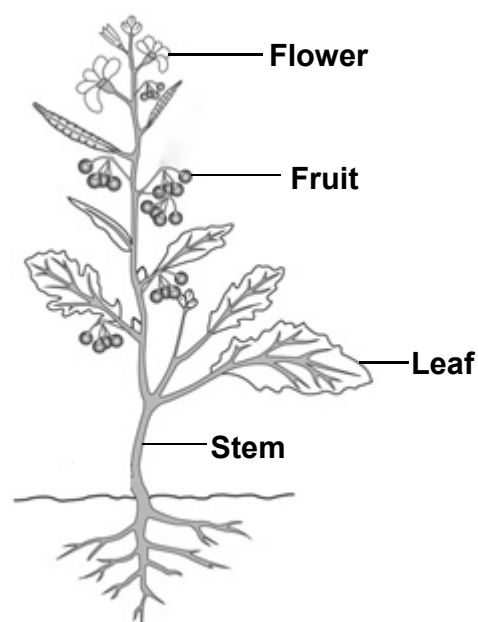


Diagram 1: A plant

2. **Diagram 2** shows the water cycle.

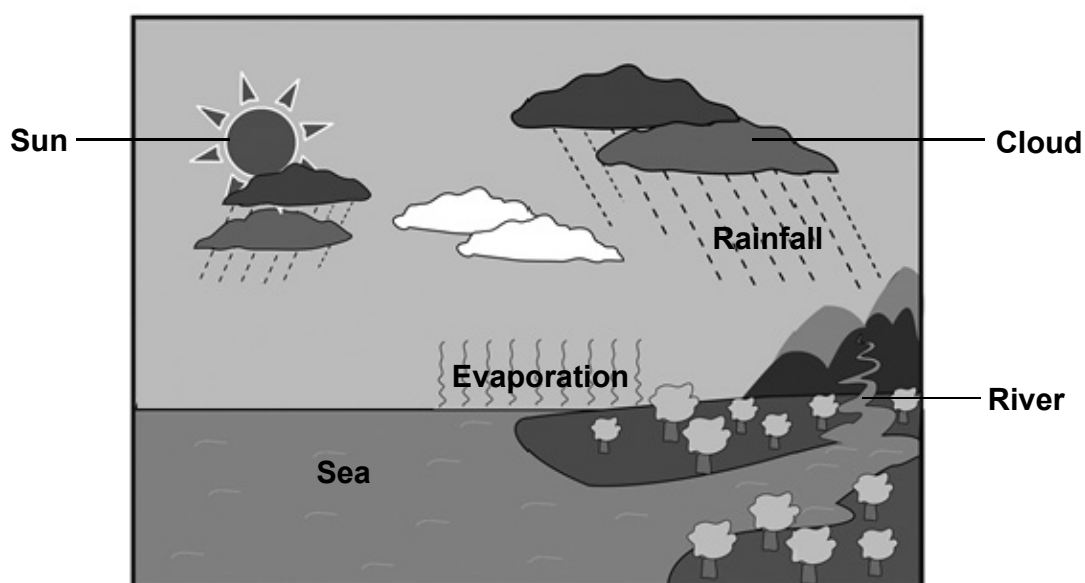


Diagram 2: The water cycle

In the water cycle, what is the **source** of energy for **evaporation** to take place?

- A Cloud
- B River
- C Sea
- D Sun

3. **Diagram 3** shows ice cubes which have been taken out of a freezer.

What is the temperature of the **ice** cubes?

- A Above 100°C
- B Between 60°C and 100°C
- C Between 25°C and 60°C
- D Below 4°C



Diagram 3: Ice cubes

4. Josh is using a torch as shown in **Diagram 4**. The torch uses a **dry cell** as a source of energy.

Which one of the following shows the **energy transformation** in the torch?

- A Chemical energy to movement energy
- B Chemical energy to light energy
- C Light energy to sound energy
- D Light energy to chemical energy



Diagram 4: Josh is using a torch

5. **Diagram 5** shows a **Cape Ground squirrel**. This animal uses its big tail to provide itself with **shade**.

Which one of the following is most probably the **habitat** of the Cape Ground squirrel?

- A Deserts
- B Rivers
- C Inside of caves
- D Dense forests



Diagram 5: A Cape Ground squirrel

Question 2 (10 marks)

Marks

- a) Write down the **main** transformation of energy taking place in pictures **1** and **2** below.

1.



Movement energy is transformed into
_____ energy.

Girl playing the guitar

2.



Chemical energy is transformed into
_____ energy.

Burning of wood

[2]

- b) Coal is one example of a fossil fuel.

Give **another** example of a fossil fuel.

[1]

- c) Fossil fuels are used to produce electricity.

Give **another use** of fossil fuels.

[1]

- d) **Diagram 6** shows Kevin who is sick. His father is feeding him.

Marks

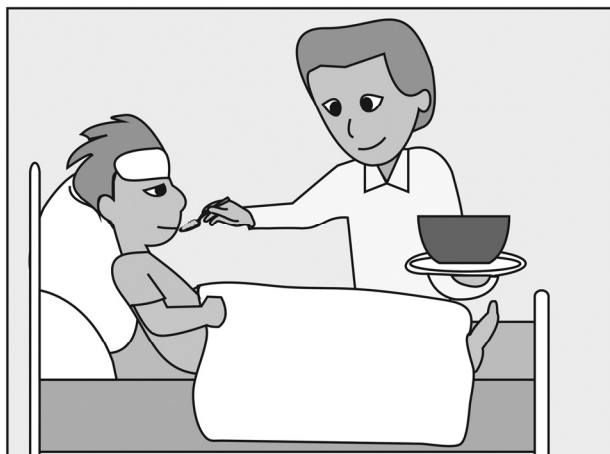


Diagram 6: Sick child and his father

It is important for Kevin to eat well when he is sick. Explain why.

[2]

- e) **Diagram 7** shows a house where electricity is being used in different rooms.

Circle two rooms in this house where electricity is being **wasted**.

[2]

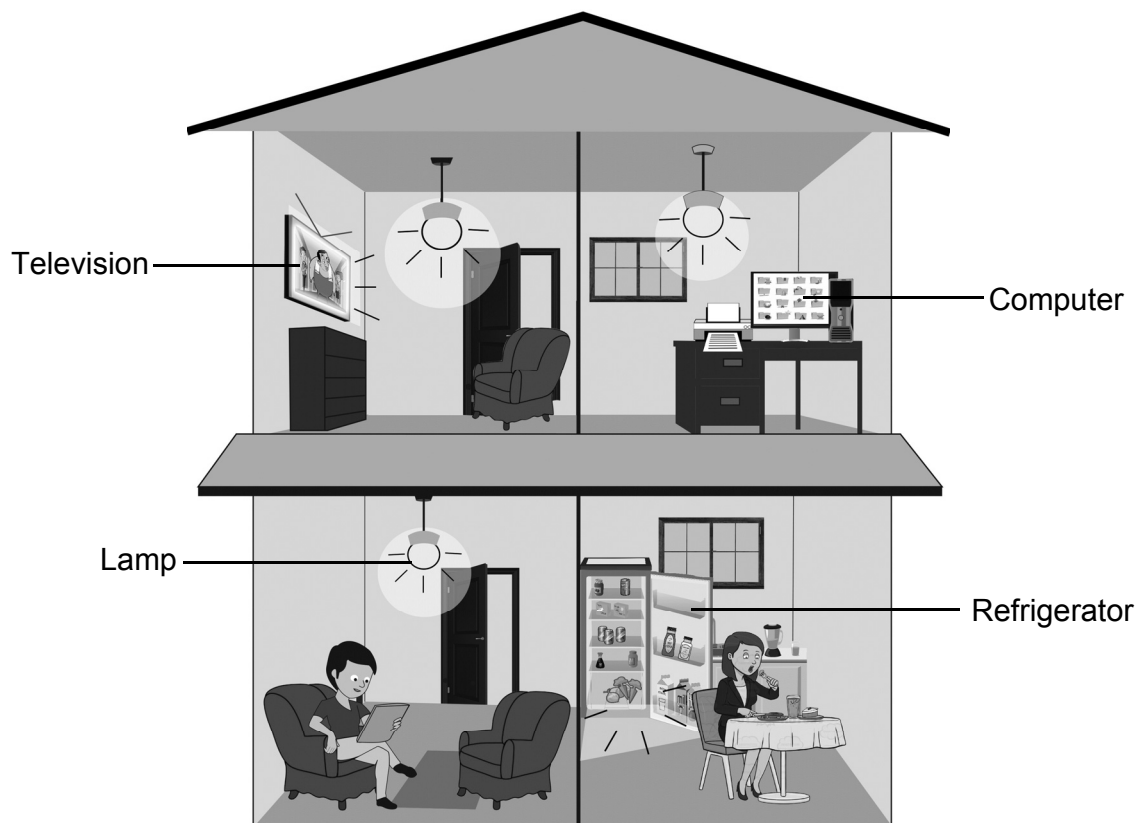


Diagram 7: Electricity being used in a house

Please turn over this page

- f) **Diagram 8** shows salt pans where seawater is evaporated to produce salt.

Marks

Salt pans



Diagram 8: Salt pans

Explain why **less** salt is produced in winter than in summer.

[2]

Question 3 (8 marks)

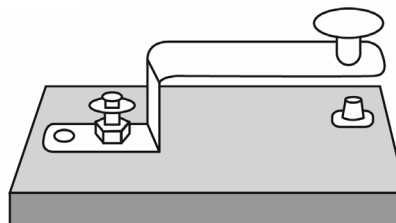
- a) Pictures **1** and **2** below show two items which are used in an **electric circuit**.

Name the items.

1.



2.



[2]

- b) **Diagram 9** shows an electric circuit which has a gap.

Marks

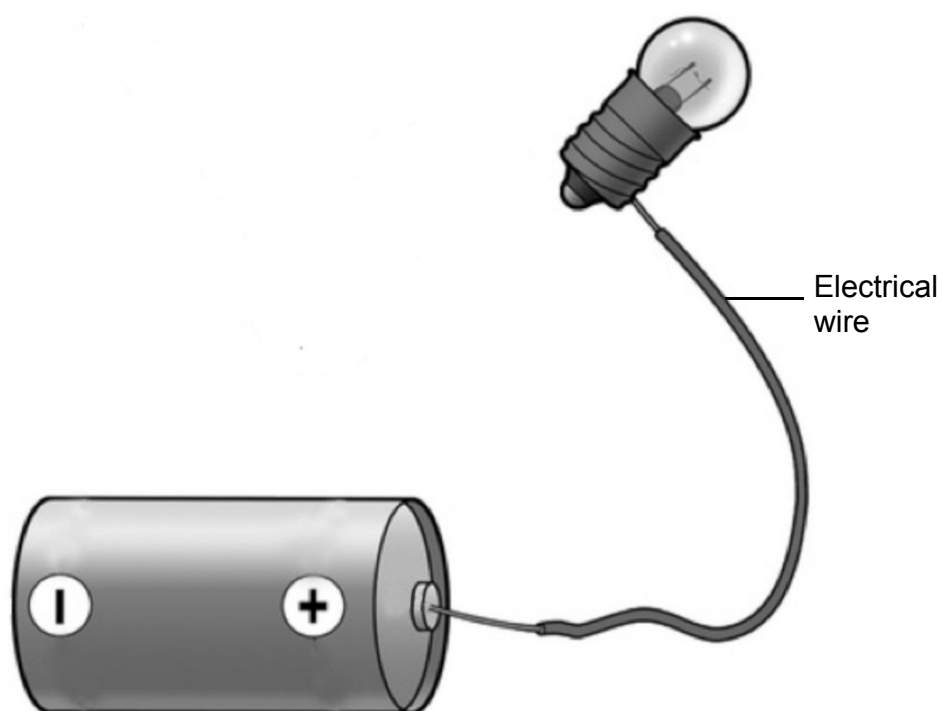


Diagram 9: Incomplete electric circuit

Complete the diagram by **drawing an electrical wire** which will **close the circuit** and will make the **bulb light up**. [2]

- c) (i) Which one of the following items could also be used to complete the circuit so that the bulb **lights up**?

- A** A rubber band
- B** A piece of cotton thread
- C** A metal ruler

[1]

- (ii) Explain your answer in c (i).

[1]

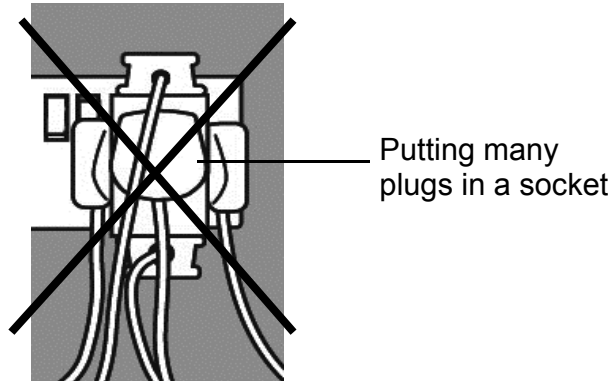
d) The pictures below show different situations involving electricity.

Marks

Put a **cross [X]** on the picture(s) showing a **dangerous** situation.

[2]

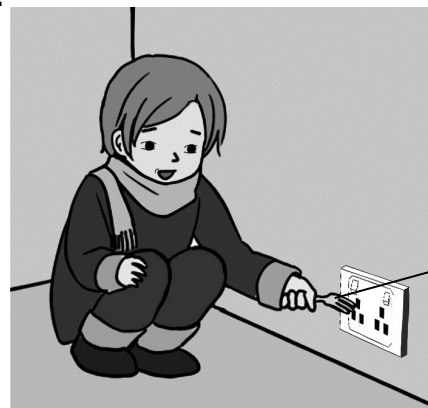
An example is given below.



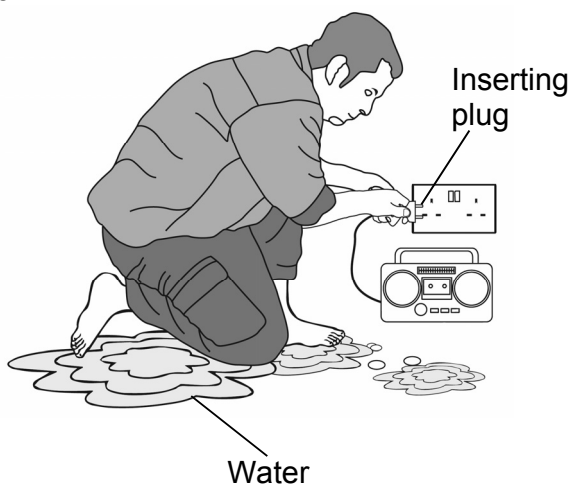
1.



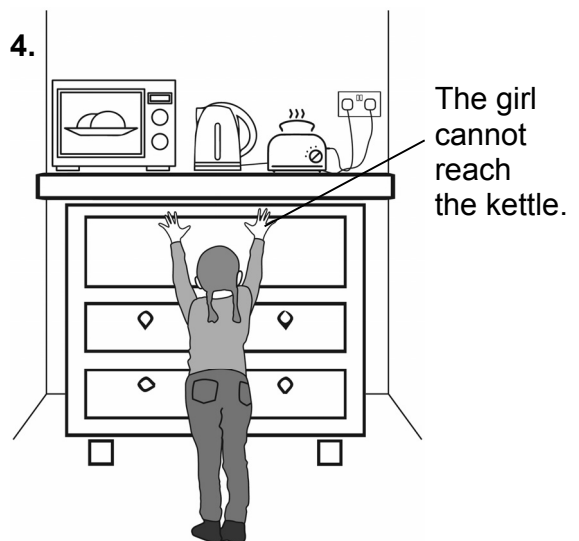
2.



3.



4.



Question 4 (11 marks)

Marks

- a) Name the process through which a **seed develops into a seedling**?

[1]

- b) Give **two common** conditions which are important for **all seeds** to grow into seedlings.

(i) _____ (ii) _____

[2]

- c) **Diagram 10** shows different stages in the transformation of a seed into a plant.

Draw stages 2 and 3.

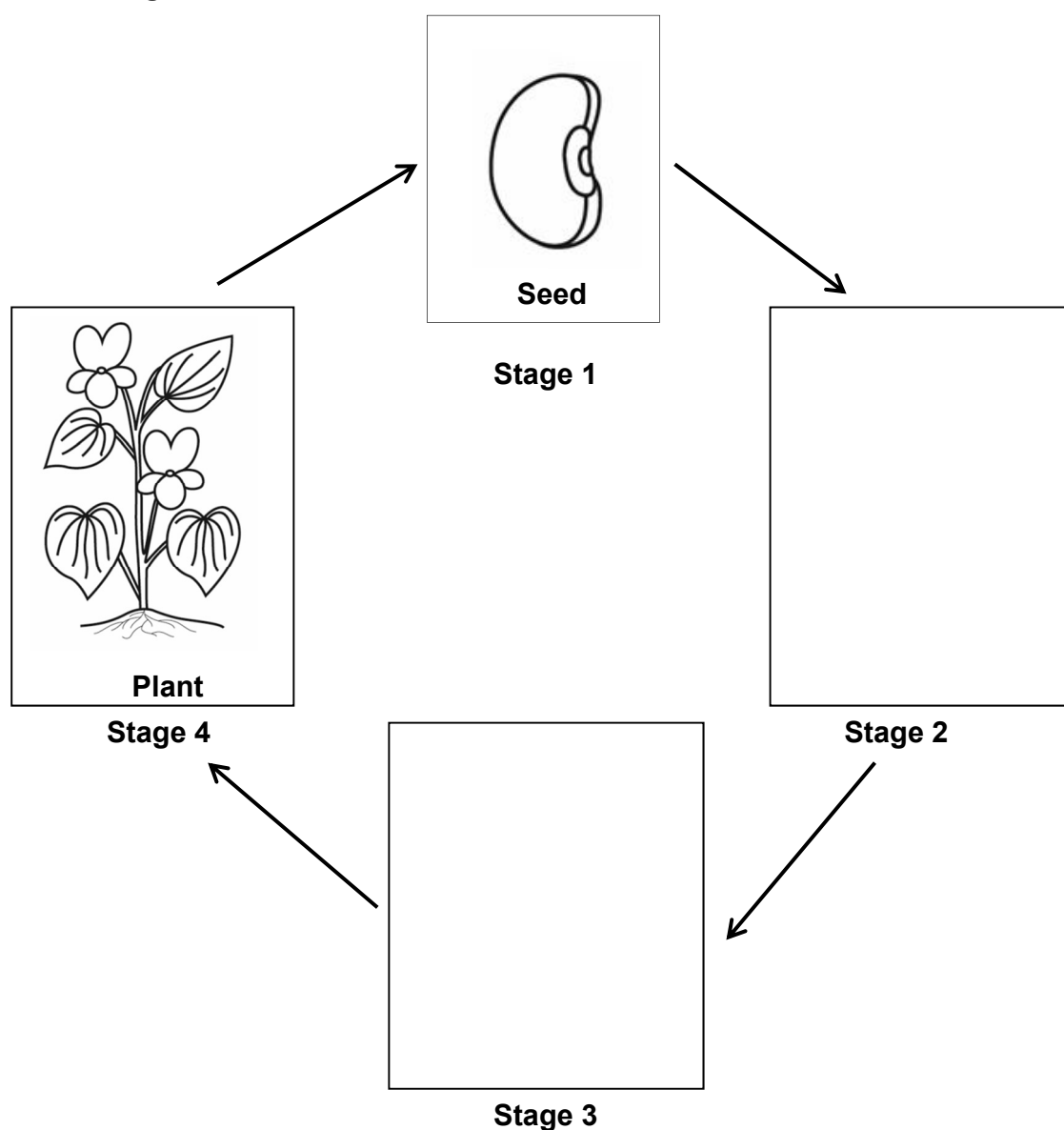


Diagram 10: Transformation of a seed into a plant

[4]

- d) **Diagram 11** shows an experiment being conducted to find out about **one of the conditions** necessary for a plant to grow well. The conditions provided to two plants, **Plant A** and **Plant B** are shown below.

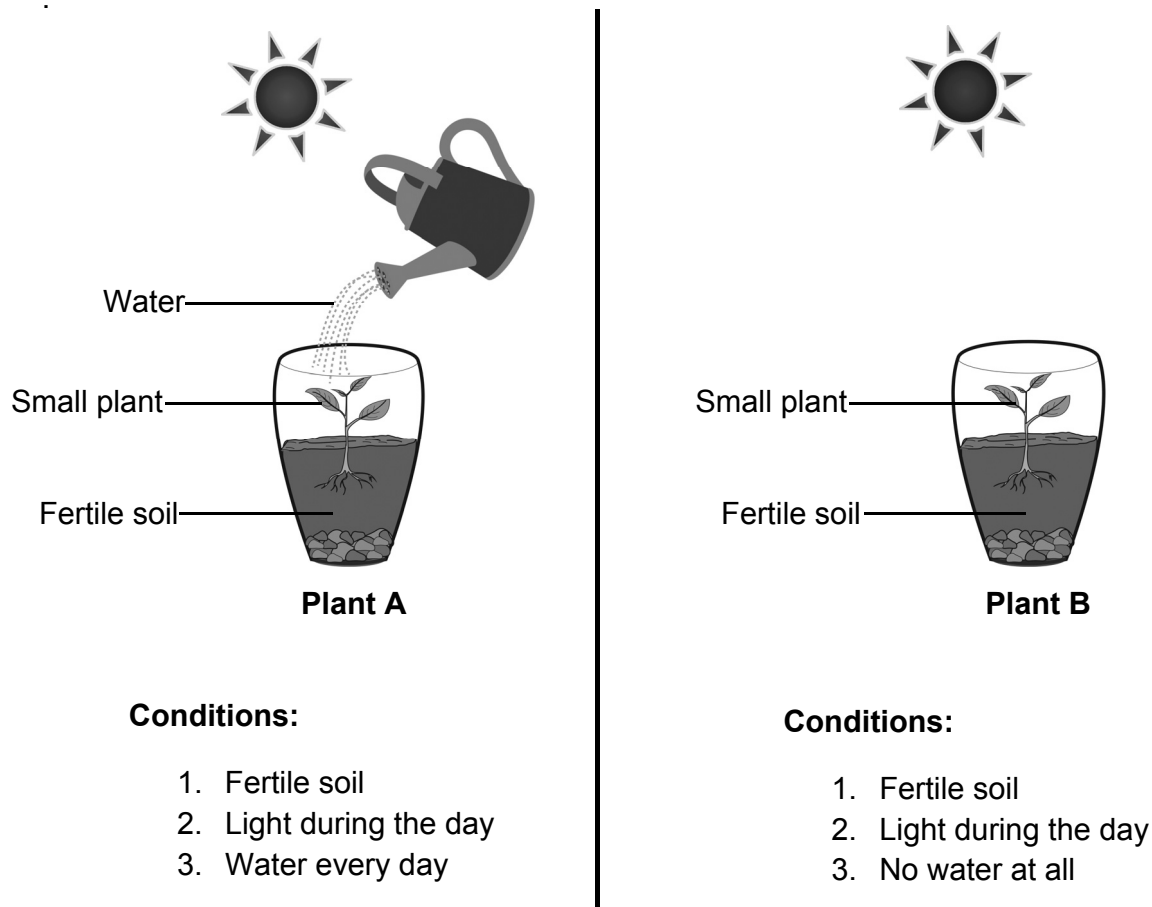


Diagram 11: Experiment on two plants

- (i) Name one **condition necessary** for a plant to grow.
- _____ [1]
- (ii) Which **condition** for plants to grow is being **investigated** in this experiment?
- _____ [1]
- e) In Rodrigues, new plants are replanted from seeds collected over the island.
- Give one reason why it is **important to replant** trees.
- _____
- _____ [2]

Question 5 (8 marks)

Marks

Diagram 12 shows a picture of Round Island which is found to the North of Mauritius.



Diagram 12: Round Island

There are no people living on Round Island **permanently**. However, many goats and rabbits were introduced on Round Island around 1840.

a) What type of animals are the **goats** and the **rabbits** to Round Island?

- A** Endangered
- B** Endemic
- C** Exotic

[1]

b) The goats and the rabbits contributed to **soil erosion** on the island.

Circle the answer which explains how.

- A** They overgrazed the land.
- B** They ate all the animals found inside the soil.
- C** They lived on the rocky parts of the island.

[1]

- c) **Diagram 13** shows the **Burrowing Boa** which once lived **only** on Round Island. The Burrowing Boa has now completely disappeared.



Diagram 13: The Burrowing Boa

An animal which has **disappeared** from Earth is said to be _____. [1]

- d) The Burrowing Boa has completely disappeared **due to soil erosion**.

(i) Which one of the following was the most likely **habitat** of the Burrowing Boa?

A The soil

B Concrete walls

C Seawater

[1]

(ii) Explain your answer in d (i).

[2]

- e) **Diagram 14** shows another type of boa, the **Keel-Scaled Boa**. It has survived on Round Island. Keel-Scaled boas live on palm trees and eat geckos (lizards).



Diagram 14: The Keel-Scaled Boa

Give one way how the Keel-Scaled Boa can be **protected**.

[2]

Question 6 (8 marks)

Marks

Sara wants to know whether **tap water** boils faster than the **bottled water** she buys.

Diagram 15 shows tap water and bottled water.



Tap water



Bottled water

Diagram 15: Tap water and bottled water

She uses two **identical** pots to boil the water.

a) Which one of the following should she **do** in her experiment?

A Use exactly the same amount of tap water and bottled water.

B Use approximately the same amount of water.

C Use more bottled water than tap water.

[1]

b) She records the temperature of the water in each pot every 3 minutes. The table below shows her recordings.

Water	Temperature of water at:				
	0 minute	3 minutes	6 minutes	9 minutes	12 minutes
Tap water	25°C	47°C	72°C	_____	95°C
Bottled water	25°C	53°C	78°C	85°C	98°C

Which one has a **higher** temperature at **12 minutes**?

Tick [✓] the correct answer.

Tap water ☐

Bottled water ☐

[1]

c) Suggest a possible temperature of the tap water at **9 minutes**.

_____ °C

[1]

- d) How can she make sure that the **measurements** of temperature she has recorded **are correct**?

[2]

- e) When she has completed the experiment, she covers each pot with a lid.
She observes droplets of water on the underside of the lids.

One pot is shown in **Diagram 16**.

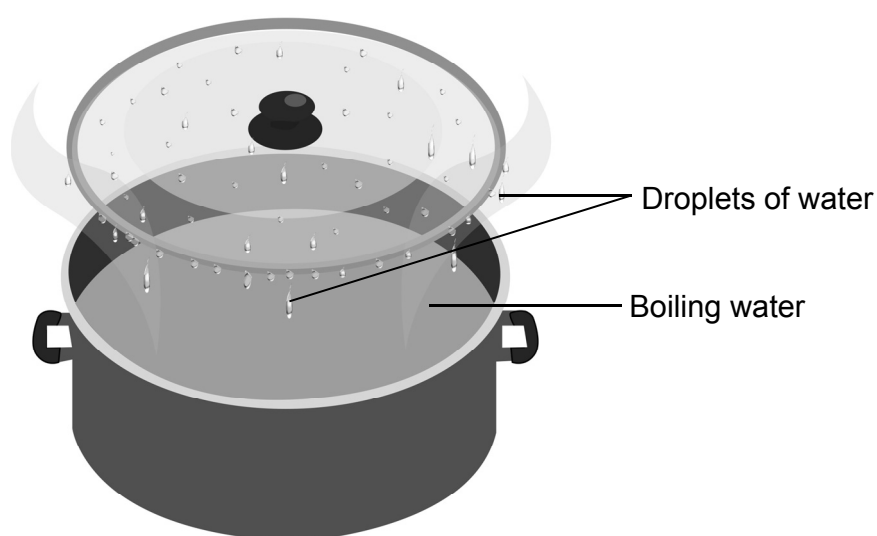


Diagram 16: Water droplets on the underside of one lid

- i) Name the **process** by which the water droplets are formed.

[1]

- ii) Give one **way how** this process is important in everyday life on the planet.

[2]

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